Harivijay R Gunaseelan

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science (BASc), Computer Engineering | 2025 - Present

- **Coursework:** Digital Computers, Numerical Methods, C++ (DSA, OOP), Embedded Systems (Arduino, sensors).

Skills

Languages: Python, C++, Embedded C++ (Arduino), MATLAB, JavaScript, C#, HTML, CSS,
AI/ML Tools: LangChain, FAISS, HuggingFace Embeddings, Scikit-learn, Pandas, RAG Pipelines, Groq API
Backend & APIs: FastAPI, Django, REST APIs, Webhooks, CORS, Cloudflare Tunnel
Hardware & Embedded: ESP32, I2C , Arduino Nano, Biomedical Sensors (SpO_2, Accelerometer)
Tools: Git, GitHub, VS Code, Render, Vercel, React, Arduino IDE, Unity, Bootstrap, CI/CD pipelines

Work Experience

Accessibility Review Team Member — **University of Waterloo**
Waterloo, ON (Remote, Project-Based) June 2026 - August 2026

- Audited **HTML source code** across **3 UW PD courses** against **WCAG accessibility standards**, identifying and documenting content quality issues impacting assistive technology users across multiple course modules.
- Produced **structured compliance reports** with cited WCAG criteria for each finding, delivering clear, verifiable, and actionable documentation that enabled stakeholders to resolve accessibility gaps efficiently.
- Aligned classification and validation standards through recurring team syncs and created structured, screen-reader-friendly transcripts translating visual course content into accessible text.

Projects

Automated Job Recommendation Platform
View Repository | *Personal Project* | View in Portfolio March 2026 - April 2026

- Built an **ML-based job recommendation system** using **Python, Pandas, and Scikit-learn** trained on **500+ real postings** across **20+ companies**, automatically matching user skillsets to relevant roles.
- Trained a **Random Forest classifier** for role matching, applied **regression** for salary prediction and **K-Means clustering** for targeting; validated performance via **80/20 train-test splitting** and accuracy scoring.
- Surfaced **top-10 job matches** with salary estimates through a **Django + Bootstrap** dashboard in under **5 seconds**.

DocuMind — AI Document Q&A Platform
View Repository | *Personal Project* | Project Link | View in Portfolio April 2026 - May 2026

- Built a **full-stack RAG application** (**React, Tailwind CSS, FastAPI**) supporting **PDF, DOCX, and XLSX** uploads, enabling natural-language queries over multi-page documents without manual searching.
- Designed an end-to-end retrieval pipeline (ingestion → chunking → **HuggingFace** embeddings → **FAISS** vector search → **Groq/LangChain** generation) returning context-aware answers grounded in uploaded document content.
- Configured recursive text splitting with **500-token chunks** and **50-token overlap** to preserve context across chunk boundaries and improve retrieval consistency for multi-paragraph document-based queries.

AI GitHub PR (Pull Request) Review Bot
View Repository | *Personal Project* | View in Portfolio April 2026 - May 2026

- Built an **AI-driven PR review bot** using **FastAPI** and **LangChain**, integrating **GitHub Webhooks** to automatically trigger code reviews on every new PR without any manual intervention.
- Fetched PR diffs via **GitHub REST API** and constructed **structured LangChain prompt chains** to route code changes through **Groq (LLaMA 3.3)**, generating categorized feedback on bugs, readability, and improvements.
- Exposed the **FastAPI backend** to GitHub's webhook delivery system using a **Cloudflare Tunnel**, enabling fully automated end-to-end PR reviews to complete within **10–15 seconds** of a pull request being opened.

Achievements & Certifications

- **FreeCodeCamp Python Developer Certification**: 300-hour certification in Python fundamentals, OOP & DSA.
- **MATLAB Onramp** (MathWorks): Certification in MATLAB syntax, matrix operations, data import and plotting.